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AliNe: A Flexible and Efficient Nextflow Pipeline for Read Alignment

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Summary

Alignment of sequencing reads against a reference genome is a fundamental step in many bioinformatics workflows. Aligner performance varies by speed, memory efficiency, and accuracy, with some tailored to specific sequencing technologies and others more versatile, which makes the selection of an appropriate tool context-dependent. To streamline this process, we present AliNe (Alignment in Nextflow), a flexible and efficient read alignment pipeline built on the Nextflow framework (Tommaso et al., 2017). AliNe contains a broad range of commonly used aligners, and is designed to accommodate any high-throughput sequencing projects.

AliNe supports short reads (both paired-end and single-end) as well as long reads generated by PacBio and Oxford Nanopore Technologies (ONT). It currently supports 16 widely used alignment tools, including BBMap (Bushnell, 2014), Bowtie (Langmead et al., 2009), Bowtie2 (Langmead & Salzberg, 2012), BWA (Li & Durbin, 2009), BWA-MEM (Li, 2013), BWA-SW (Li & Durbin, 2010), GraphMap2 (Marić et al., 2019), HISAT2 (Kim et al., 2019), Kallisto (Bray et al., 2016), Minimap2 (Li, 2021), ngmlr (Sedlazeck et al., 2018), novoAlign (*novoAlign / Novocraft*, n.d.), nucmer (Marçais et al., 2018), STAR (single or two-pass mode) (Dobin et al., 2013), subread (Liao et al., 2013) and sublong (Liao et al., 2013). These aligners are integrated into a single, easy-to-use workflow, providing a unified entry point for any project requiring alignment and giving users the flexibility to choose the best tool for their specific data and objectives. AliNe is designed to minimize user inputs and avoid common parameter mistakes (e.g. scoring system, strandedness, orientation).

Statement of Need

The complexity of sequencing data analysis, especially with the advent of next-generation sequencing (NGS), has created a demand for robust, scalable, and reproducible workflows that can handle large datasets while allowing users to flexibly choose their preferred analysis tools. Numerous bioinformatics analyses rely on read alignment steps that can play a major role in the quality of the downstream results. Each alignment tool may behave differently and have its own strengths. Users may need to try out several alignment tools in order to achieve optimal results specific to a project's requirements. AliNe empowers researchers with a single-entry point for alignment, eliminating the need to juggle between disparate tools. By integrating multiple aligners into a unified pipeline, AliNe simplifies the process of alignment algorithm selection and establishes a standardized workflow that ensures both reproducibility and transparency across analyses

The majority of pipelines integrating read alignment steps are specialized for particular sequencing technologies and types of analyses (Riesgo-Ferreiro et al., 2022) (Patel et al., 2024) (Garcia

et al., 2024) (Chafin et al., 2024). There remains a lack of unified, versatile workflows capable of integrating aligners into a single, user-friendly platform. AliNe addresses this challenge by providing a comprehensive, streamlined solution for alignment tasks across a wide range of sequencing technologies, whatever the field of application. Moreover, AliNe can be integrated as a sub-workflow into other Nextflow pipeline and provide a standardize step that offer users a wider choice of alignment tools.

Workflow Overview

AliNe takes sequencing reads in FASTQ format and aligns them to a user-supplied reference genome in FASTA format, using user-selected aligners. The workflow is fully configurable to accommodate various data types, and gives users the ability to fine-tune tool alignment parameters independently. Nevetheless, AliNe is designed to minimize user inputs and avoid common mistakes like the definition of scoring system used by the sequencing machine (Phred+33, Phred+64 and Solexa), and for RNAseq data the strandedness (stranded or unstranded) or the orientation in case of paired-end reads (forward reverse i.e., fr; reverse forward i.e., rf; forward forward i.e., ff). This information may impact the results. If not provided by the user, AliNe will deduce this information and adapt the option of the different downstream aligners accordingly.AliNe accompanies also the user to select proper aligner depending data input provided.

To run AliNe you need to select one or several aligners, and tell what type of data you provide as input: short single-end reads, short paired-end reads, long reads generated by PacBio or long reads generated by Oxford Nanopore Technologies (ONT). AliNe integrates an optional read trimming step using Fastp (Chen, 2023). It also includes optional quality control at every stage with FastQC (*FastQC*, 2015). For RNA-seq data, the pipeline automatically detects the library type (strandedness and orientation) via Salmon (Patro et al., 2017) and configures the appropriate parameters for downstream alignment tools accordingly. Additionally, AliNe uses SeqKit (Shen et al., 2016) to detect the sequence quality scoring system and standardizes it to Phred+33 if necessary. Once preprocessing is complete, alignment is performed using the aligners selected by the user. Finally, with the help of Samtools (Danecek et al., 2021), AliNe outputs aligned reads in sorted BAM format, ready for downstream analyses such as variant calling, gene expression quantification, or structural variation detection. A final stage will resume all FastQC analyses, and log output from compatible aligners using MultiQC (Ewels et al., 2016). All these steps are summarized in Figure 1.

The pipeline supports parallel execution, ensuring efficient processing of large datasets by distributing tasks across available CPU cores. With Nextflow's inherent support for reproducibility, users can confidently rerun analyses with consistent results.

The pipeline is modular, enabling future expansion and the integration of additional tools or customization of downstream analysis steps. Users can adapt AliNe to their specific research needs, whether for RNA-seq, ChIP-seq, or whole-genome resequencing projects.

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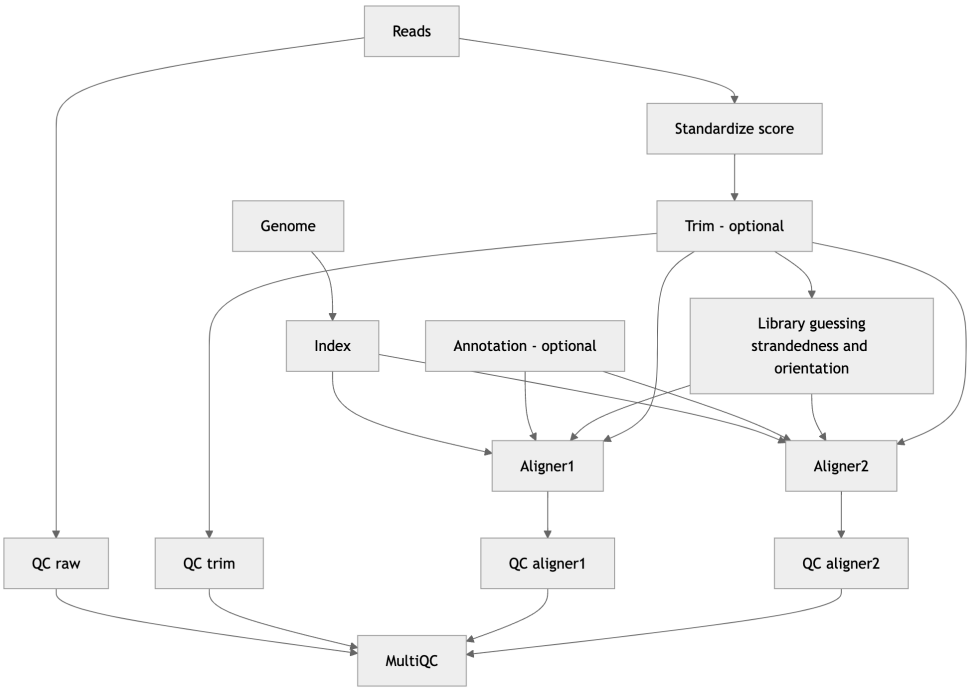

Figure 1: Overview of the different steps inplemented in AliNe.

Availability

AliNe is freely available under an open-source license. The source code, along with detailed documentation and examples, can be found at https://github.com/Juke34/AliNe. Contributions and community feedback are welcome.

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